

School of Science, Computing and Engineering Technologies



COS10025

Technology in an Indigenous Context Project

Innovation Concept Report

Project Title: Sustainable Water Solutions for Tua Chua Student

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Part A: Overview

1.1 Project Overview:

This project aims to address the ongoing water scarcity in Tủa Chùa District, Điện Biên Province, where many households lack reliable access to clean drinking water. The project proposes five design ideas focused on improving water collection, purification, storage, and distribution for local communities. These solutions explore approaches such as solar-powered purification systems, rainwater harvesting, and improved water storage methods. Together, they aim to provide safe drinking water, especially for remote villages without centralized water infrastructure, while promoting sustainable and climate-resilient water management.

1.2 Identified Challenge:

The main challenge in Tua Chua is the chronic shortage of clean and reliable water, particularly during the dry season when natural water sources often dry up. The district's karst terrain, scattered settlements, and limited infrastructure make it difficult to build stable water supply systems, leaving many communities dependent on unsafe or seasonal water sources.

Part B: Design Ideas

2.1 Design Idea 1: The "Highland Mist" Hybrid Water System:

Description of the Design Idea

The "Highland Mist" Hybrid Water System is designed to address water scarcity in mountainous karst regions, where rainfall is high, but surface water is limited because rain quickly drains underground. As a result, many communities struggle to access clean water.

The system collects water from atmospheric moisture instead of relying on groundwater. Fog is captured using a 3D Raschel mesh and purified through a bio-sand filtration system to remove pathogens and impurities. A LoRaWAN-based IoT network monitors water levels in real time and provides reservoir information. By combining fog harvesting, natural filtration, and digital monitoring, the system provides a reliable and sustainable water supply.

Digital Drawing of Design Idea

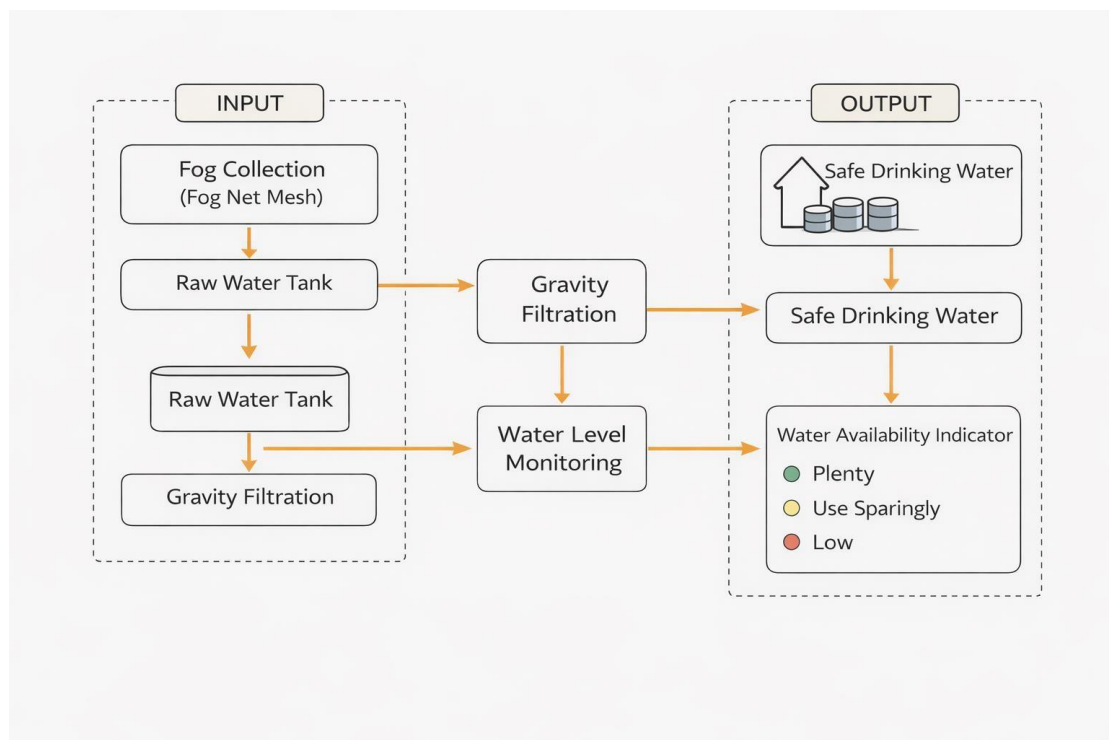


Figure 1: Flowchart of the fog water collection and filtration system.

Design Specifications

Approach 1: Atmospheric Water Collection (The Source)

Technology: 3D Raschel Mesh

The system uses a double-layer Raschel mesh to capture fog droplets more effectively. This design helps collect up to 300% more water than traditional flat mesh nets (Qadir et al., 2018).

Devices:

Community Fog Wall: A modular fog-harvesting structure of about 40 m² installed on mountain ridges to collect large amounts of fog water for community use.

Home-Mist Unit: A smaller 4 m² device installed near households so remote families can also access clean water.

Approach 2: Natural Purification (The Safety) Technology: Bio-sand Filtration

Bio-sand filtration is a passive purification method using a biological layer called *Schmutzdecke*, which removes pathogens while sand layers trap sediments and impurities (Sobsey et al., 2008).

Devices:

1. Village Cistern: A 10,000-liter ferro-cement tank with integrated sand filtration to purify water before distribution.
2. Household Clay Filter: A small ceramic filter that provides additional purification for drinking water.

Approach 3: Resource Management (The Sustainability)

Technology: LoRaWAN: enables long-distance, low-energy data transmission (5 to 10 km) in rural areas (Behailu et al., 2020).

Devices:

1. Tank Sentry: A solar-powered IoT sensor that monitors water levels.
2. Status Board: A community display showing water availability with LED indicators.

Hardware and Software Requirements

1. Hardware Components

3D Raschel Mesh: UV-resistant polypropylene mesh used to capture fog droplets.

Heltec LoRa V3 Boards: Low-power microcontrollers used for LoRa communication and data transmission.

Ultrasonic Sensors (JSN-SR04T): Waterproof sensors used to measure water levels in the storage tank.

Power System: Small solar panels combined with 18650 Li-ion batteries to support continuous system operation.

Structural Materials: Treated bamboo or galvanized steel frames and ferro-cement storage tanks for durability and affordability.

2. Software Requirements

SketchUp: Used to design the structure and simulate the stability of fog-collection frames.

Blynk IoT Platform: A mobile dashboard that allows community leaders to monitor water levels and system performance in real time.

How does your design idea work/function?

The FHS-SRM system operates through three main stages: fog collection, water purification, and monitoring. **1. Fog Water Collection**

Fog carried by wind passes through the 3D Raschel mesh installed in elevated areas. Tiny droplets stick to the mesh fibers, combine into larger drops, and fall into PVC gutters due to gravity. The collected water is then directed to a storage tank (Qadir et al., 2018).

2. Water Purification

The harvested water flows into the Village Cistern and passes through a bio-sand filtration system. A biological layer called *Schmutzdecke* removes pathogens while sand layers filter out sediments and other impurities (Sobsey et al., 2008).

3. Water Monitoring and Management

The water level in the tank is monitored by the “Tank Sentry,” a solar-powered IoT device with ultrasonic sensors. Data is transmitted through a LoRaWAN network and displayed on the “Status Board.” If the water level drops below a critical threshold, the system sends an alert so that the community can manage water use effectively (Behailu et al., 2020).

Benefits of the Design Idea

The system offers several advantages.

1. The fog-harvesting mesh is designed to remain efficient in high-altitude wind conditions.
2. The system uses gravity-based filtration and solar-powered electronics, which minimizes energy consumption and environmental impact.
3. The modular design allows the system to expand easily as community water demand increases.

Community Impact

1. The system can significantly improve public health by reducing waterborne diseases through bio-sand filtration (Sobsey et al., 2008).
2. It also reduces the need for expensive water transportation, helping communities become more economically independent.
3. In addition, the monitoring system provides data that allows local leaders to manage water resources more effectively.

Cultural Appropriateness and Guidelines

The design aligns with H'mong and Thai communities by using familiar materials and natural filtration methods.

1. It supports **Access & Equity** with household units for elderly or disabled residents.
2. Using ferro-cement and local bamboo improves **Affordability**, reducing costs by about 40% compared to industrial systems.

Constraints of the Design Idea

Despite its benefits, the system has some limitations.

1. Fog density can vary depending on seasonal weather conditions, which may reduce water collection during dry periods (Qadir et al., 2018).
2. In addition, maintaining the IoT monitoring system requires basic technical knowledge. If trained personnel are not available, the digital monitoring component may become difficult to maintain (Behailu et al., 2020).

2.2 Design Idea 2: SpringGuard Gravity Water System

Description of the Design Idea

The SpringGuard Gravity Water System ensures the availability of clean and pure water for the mountainous region of Tà Chùa, Điện Biên, Vietnam. Most people source water from natural springs, which are prone to contamination from animal waste, soil, and other external factors. Ineffective water protection and sanitation are responsible for waterborne diseases in developing countries (Fewtrell et al., 2005).

The SpringGuard Gravity Water System ensures effective water protection at the source, and gravity-fed water delivery is used for distribution. A sealed concrete spring box at the outlet ensures that the water is not contaminated. Gravity-fed water is delivered through a PVC pipe, and a cloth-sand filter is placed at the community tap stand for improved water clarity. It is easy to build and uses cost-effective materials, making it sustainable for long-term use.

Digital Drawing of Design Idea

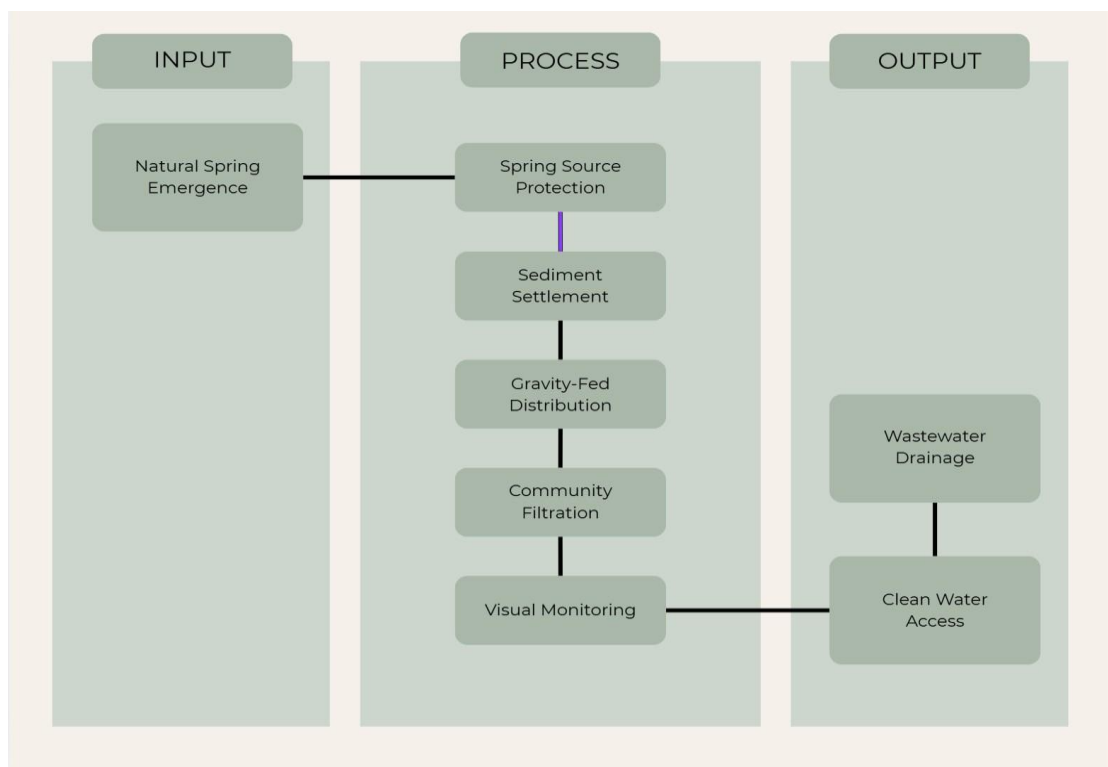


Figure 2: Functional Process Flow of the SpringGuard Source-Protection Water System

Design Specifications

The system includes three main components.

Technology: Spring Protection Structure Devices

1. Sealed concrete spring box
2. Internal sediment chamber for particle settling
3. Overflow outlet for pressure regulation

Technology: Water Distribution System Devices

1. Gravity-fed PVC pipeline transporting water downhill
2. Buried pipe sections to reduce damage and temperature fluctuation
3. Inspection points for maintenance (Kleiner, 2001)

Technology: Community Collection Point

Devices

1. Tap stand with cloth–sand filtration
2. Elevated concrete platform
3. Drainage channel to prevent stagnant water

This configuration protects the water source while requiring minimal technology and maintenance.

Approach 1: Source Protection

Protecting water directly at the spring prevents contamination before it reaches users. Sealing the spring box isolates water from soil, animals, and runoff, which are major sources of microbial contamination in rural water supplies (Luby et al., 2008).

Approach 2: Gravity-Based Distribution

The system uses natural gravity to transport water downhill instead of pumps or mechanical systems. This eliminates the need for electricity, fuel, or complex equipment, making it suitable for isolated mountainous communities.

Approach 3: Community-Level Filtration

At the tap stand, a cloth and sand filter removes debris and fine particles before water collection. Sand filtration systems have been shown to reduce bacteria such as *E. coli* in drinking water (Elliott et al., 2008).

Hardware and Software Requirements

1. Hardware

Concrete and cement for the spring box

PVC pipes and connectors

Sand and cloth for filtration

Drainage pipes and tap fixtures

Basic construction tools

2. Software

No digital software is required for daily operation. For community use, maintenance guidelines and inspection procedures are crucial. And without the presence of complex technology, the system is user-friendly and easy to maintain in rural areas.

How the Design Idea Works

The SpringGuard system operates through a simple process.

First, natural spring water enters the sealed concrete spring box to prevent contamination while allowing sediment to settle. Clean water then flows into a PVC pipeline connected to the spring box outlet. Because the spring is located on a higher floor, gravity will push the water downhill toward the village naturally.

Finally, water reaches the community tap stand where it passes through a cloth–sand filtration layer. The cloth traps larger debris, while fine sand holds smaller particles and some microorganisms. Finally, villagers collect water from the tap in a hygienic environment. This process ensures continuous water flow and provides basic purification by using natural resources.

Benefits of the Design Idea

The SpringGuard system has several benefits.

1. Improve public health: protecting the water source reduces exposure to microbial contamination and waterborne diseases (Fewtrell et al., 2005).
2. Low operation cost: electricity, fuel, or mechanical equipment is not needed for the gravity-fed system.
3. Environmental protection: spring protection helps stabilize the surrounding soil and reduce erosion.
4. Long-term sustainability: Durable materials and simple maintenance practices support long-term operation (Kleiner, 2001).

Community Impact

1. Reliable access to safe water greatly improves the quality of daily life in rural communities. When water is available nearby, households will spend less time traveling great distances for it. Particularly women and children
2. Community participation in maintaining the system can strengthen shared responsibility and cooperation among people. Local construction and maintenance also allow residents to contribute labor and skills, increasing community ownership of the system. Research shows that community participation is a key factor in the sustainability of rural water projects (Marks et al., 2014).

Cultural Appropriateness and Guidelines

1. The SpringGuard system is designed for local practical use. The tap stand illustrates the tradition of sharing and collecting methods commonly used in rural areas. Also, the materials such as sand, cloth, and PVC pipes are familiar to the locals, making the work flow among workers operate smoothly.
2. Community involvement in system management also aligns with collective decision-making traditions. The locals can implement maintenance schedules and inspection guidelines to ensure the functionality of the system in the long term.

Constraints of the Design Idea

Despite its benefits, the system has limitations.

1. The design depends heavily on perennial spring, and water flow can be reduced during dry periods. Construction in mountainous areas may require additional time and funds for transporting materials and installing infrastructure.
2. The project also needs consistent maintenance, regular inspection, and cleaning of filtration layers from the community to ensure long-term success.

2.3 Design Idea 3: Rainwater harvesting system integrating Solar-powered disinfection

Design Description

The smart rainwater harvesting system integrates rooftop rainwater collection, multistage filtration, and solar-powered disinfection into a single sustainable solution. This system innovates by incorporating an automatic first-flush diversion mechanism to remove initial contaminants and energy-efficient treatment technologies to ensure safe and economical drinking water.

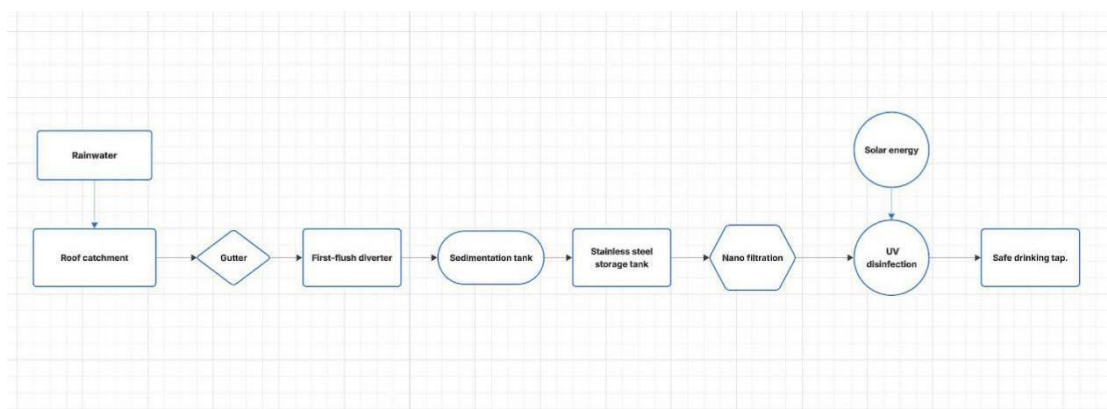


Figure 3: Flowchart of Smart Solar-Powered Rainwater Harvesting System

Design Specifications

Approach 1: Physical Filtration: The water is filtered to remove leaves, soil, and other impurities before it enters the storage tank.

Devices

Mesh Screen Filter: It removes large debris such as leaves, twigs, and other solid particles from rainwater.

Sand and Gravel Filter: It removes dirt and small debris

Approach 2: Automated First-Flush Diversion System:

Technology: First-flow Polluted Diversion: This system manages pollutants in the initial phase of rooftop stormwater runoff by diverting the most contaminated rainwater before clean water enters the storage tank. It effectively removes suspended solids, pollutants, and heavy metals (Martinson & Thomas, 2014; Syarifah et al., 2024).

Devices

Calm-Inlet Pipe (U-shaped): Reduces turbulence when water enters the tank and prevents sediment disturbance.

First-Flush Diverter: A first-flush diverter automatically diverts the initial runoff that contains the highest concentration of pollutants away from the storage system.

Approach 3: Sterilization:

Technology: Solar-Powered Disinfection: The method applies solar water sterilization technology, in which water is sterilized using ultraviolet (UV) light to eliminate bacteria before being used as safe drinking water for humans. Solar water sterilization is a suitable and cost-effective technology for household water treatment (García-Gil et al., 2021) in resource-limited areas.

Devices

Solar-Powered UV Disinfection: Uses ultraviolet radiation to inactivate microorganisms without chemicals and operates independently from the grid.

Solar Photovoltaic (PV) Panel: A solar photovoltaic panel generates renewable electricity from sunlight to power the UV disinfection unit, enabling the system to operate in off-grid or remote areas.

Hardware and Software Requirements

Hardware Requirements

Rainwater collective roof: The utilization of existing roof structure reduces installation costs and enhances local economic viability.

PVC Pipe: PVC pipe is used for transporting water between different stages.

First-flush Diverter: The first-flush diverter removes impurities and dirt on the roof surface. Consequently, the quality of water entering the system is entirely enhanced before entering the next stage of filtration and sedimentation.

Sedimentation Tank: It allows for the settlement of denser impurities and sediments, minimizes filter component wear and tear, extending filter life.

Stainless Steel Storage Tank: Stainless steel tanks are used because they are durable, have a long lifespan, and are easy to clean periodically, with their corrosion resistance and low scale formation (Manga et al., 2022).

Nanofiltration Cartridge: The nanofiltration cartridge can decrease the level of heavy metals and enhance the turbidity level of the water (Jadhao et al., 2024)

UV Disinfection Chamber: This section ensures that the water is disinfected using ultraviolet light, eliminating the need to use chemicals in the process.

Solar Panel System: This section provides the ultraviolet sensors with the required energy, hence allowing the system to function independently without relying on the national grid.

Software Requirements

Automated control of UV activation based on water flow and system status

Triggering alerts for low water levels and scheduled maintenance

How does the design idea work

Rainwater is collected on the roof surface and channeled to a first-flush diverter, where the contaminated water is separated. After that, water is passed to a sedimentation tank, in which heavier contaminants and solids settle at the bottom of the tank. Then, the water is channeled into a stainless steel storage tank without disturbing the settled contaminants. The stored water is filtered using a nano-filter and sterilized before being used by a solar-powered ultraviolet sterilization chamber. Moreover, solar power is used to generate electricity during the day, which is stored and used to power the ultraviolet sterilization unit.

Benefits

- 1.It provides a reliable resource for safe drinking water from a natural resource.
2. It can reduce the time taken for water from far sources.
3. Solar energy makes the system operate independently from the national electricity network, making it a solution for remote areas.

Community Impact

The design helps to enhance the quality of domestic water. It also helps in reducing the financial burden on poor households, who would spend their resources on buying water or the treatment of water-borne diseases. Additionally, the storage capacity helps in building community resilience during dry seasons.

Cultural Suitability and Guidelines

1. **Access and Equity:** The proposed system will provide access to clean water for the local community without a long-distance process.
2. **Health Safety:** The proposed system produces safe water for locals.
3. **Environmental Health:** The system reduces the environmental hazards of overexploitation of groundwater resources.

Constraints of the Design Idea

This system has a reliance on rainfall and is unable to satisfy demand in the case of droughts if the storage capacity is small. Moreover, the cost of the initial investment in the high-technological equipment is quite high. Therefore, financial assistance may be required from the local government or development agencies during the initial phase of implementation.

2.4 Design Idea 4: The “Limestone Lens” Adaptive Storage System

Description of the Design Idea

The Limestone Lens is a decentralized rainwater harvesting network engineered for geologically sensitive karst landscapes. By leveraging an underground ferro-cement application, the storage system maximizes land utility and effectively reduces rapid evaporation. Following the Multi-Barrier Principle (Dinka, 2018), it ensures water safety through multiple procedures of sterilization. This architecture achieves energy and operational autonomy using solar arrays and sand batteries, while IoT systems enable real-time water-level monitoring.

Digital Drawing of Design Idea

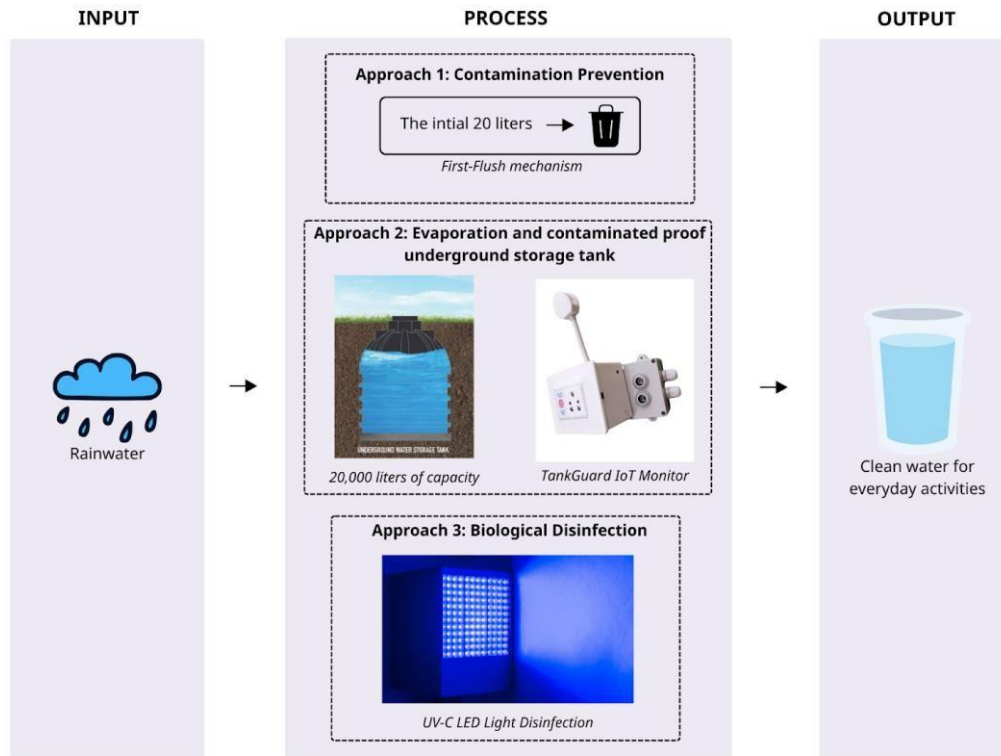


Figure 4: Flowchart of the Limestone Lens adaptive storage system

Design Specifications

To implement the Limestone Lens system effectively, the approaches must be strategic to deal with geographically challenging area and ensure water quality.

Approach 1: Contamination Prevention (Prevent)

This first approach acts as a gatekeeper to filter the unwanted waste out of the storage system, significantly improving water quality.

Technology: First-Flush Diverter System.

Devices:

Rain-Splitter: A vertical-placed mechanical pipe that captures the first 20L of unwanted waste and diverts it away from the storage tank.

Smart Contamination Alert Sensor: An IoT turbidity sensor that monitors the water's purity and alerts users if it exceeds safe limits.

Approach 2: Evaporation Reduction & Safe Storage (Reduce)

Once the water is cleaned of surface debris, it needs a safe place to sit without rapid evaporation or getting contaminated. The goal is to keep the water cool and block direct sunlight exposure to avoid algae growth.

Technology: Underground Ferro-Cement Storage.

Devices:

Geo-Vault: A 20,000L underground tank made of basket-shaped resilient ferro-cement to handle the physical pressure of karst terrain.

TankGuard IoT Monitor: A sensor that tracks water levels in real-time and sends the data to a village dashboard via LoRaWAN. **Approach 3: Biological Disinfection (Protect)**

The final approach is the last safety procedure to kill invisible germs, ensuring water is biologically safe by using UV LED sterilization.

Technology: Solar-Powered UV-C LED Disinfection.

Devices:

Solar-Purity Wand: A device at the tap that uses UV-C light to neutralize bacteria and pathogens without using chemicals.

EnergyVault Module: A solar-powered generator using sand batteries and supercapacitors to store energy, ensuring its self-sustainability.

Hardware and software components

Figure 5: Estimated implementation cost for the Limestone Lens adaptive storage system

Component	Category	Purpose	Estimated Cost (USD)
Ferrocement Materials	Hardware	Material of the 20,000L GeoVault.	\$500.00 – \$650.00
ESP32 LoRaWAN Node	Hardware	Central microcontroller for data processing and transmission.	\$25.00 – \$40.00

UV-C LED Array	Hardware	Biological disinfection	\$60.00 – \$100.00
JSN-SR04T Sensor	Hardware	Waterproof ultrasonic sensor for waterlevel monitoring.	\$12.00 – \$18.00
SEN0189 Sensor	Hardware	Turbidity sensor for inlet water clarity monitoring.	\$10.00 – \$15.00
Solar & Sand Battery	Hardware	50W Solar panel and thermal sand storage components.	\$150.00 – \$220.00
Rain-Splitter Parts	Hardware	PVC piping, floating ball valve, and debris mesh.	\$40.00 – \$60.00
Arduino/ESP32 IDE	Software	Firmware development and logic implementation.	Free
LoRaWAN Stack/Dashboard	Software	Data transmission and village monitoring interface.	\$0.00 – \$10.00/month
Estimated Total			\$812.00 – \$1,138.00

How it works/ functions

Rainwater is collected from household rooftops. To ensure safety, the Limestone Lens system uses a first-flush device called the Rain-Splitter. This device diverts the first 20 liters of contaminated rainwater away from the storage container. A turbidity sensor also monitors the water and indicates if the collected rainwater quality is compromised.

After the first 20 liters are diverted, a ball rises to close the valve of the Rain-Splitter, allowing cleaner rainwater to flow into a 20,000 L underground ferro-cement GeoVault. This underground storage is suitable for limestone regions because it maintains a cool temperature, prevents algae growth, and reduces evaporation (Helmreich & Horn, 2009). The tank also contains the TankGuard IoT monitor, which uses ultrasonic sensors and LoRaWAN to measure water levels in real time (Jabbar et al., 2024; Mohammed et al., 2019).

For disinfection, the Solar-Purity Wand device treats the collected rainwater during use. Water from the middle layer of the Geo-Vault passes through UV-C LEDs powered by solar energy, eliminating the need for chemical disinfectants (Song et al., 2016). This approach follows the Multi-Barrier Water Safety Principle (Dinka, 2018) and allows the system to operate independently using renewable energy.

Benefits

Safety: The multi-barrier approach ensures high-quality potable water by removing physical, chemical, and biological contaminants.

Self-sustainability: Sand batteries provide energy independence, ensuring the system remains operational during extended dry seasons.

Affordability: The use of ferro-cement material is more cost-effective per liter than the average plastic tanks and utilizes local construction skills.

Community Impact

Health: Significantly reduces the risk of water-borne diseases through chemical-free sterilization, effectively allowing residents to focus primarily on their education, physical and mental well-being.

Livelihoods: Encourages local training in masonry and IoT maintenance, creating sustainable technical jobs.

Cultural Appropriateness: The design utilizes community-level management and localized construction materials (sand/cement) rather than alien, centralized architecture.

Constraints of the Design Idea

Initial Cost: Initial material costs for a 20,000L tank and IoT hardware remain as an invisible barrier for low-income households.

Technical Literacy: Maintenance of IoT nodes and sand batteries requires specialized community training for incident response and to prevent system failure.

2.5 Design Idea 5: Double Filtration and Smart Warning System

Design Description

The innovation of this system is Double Filtration with Smart Warning by means of Fuzzy Set Theory, which assesses spatial and temporal risks in water and locates the tanks away from areas of chronic pollution, such as disposal sites. This system will deal with transitory flood pollution through physical separation. The sequence of processes will be: Fuzzy Set, Turbidity Sensor, Smart Solenoid Valve, Double Filtration by means of quartz sand and activated carbon, Traffic Light Indicator System, and then Safe.

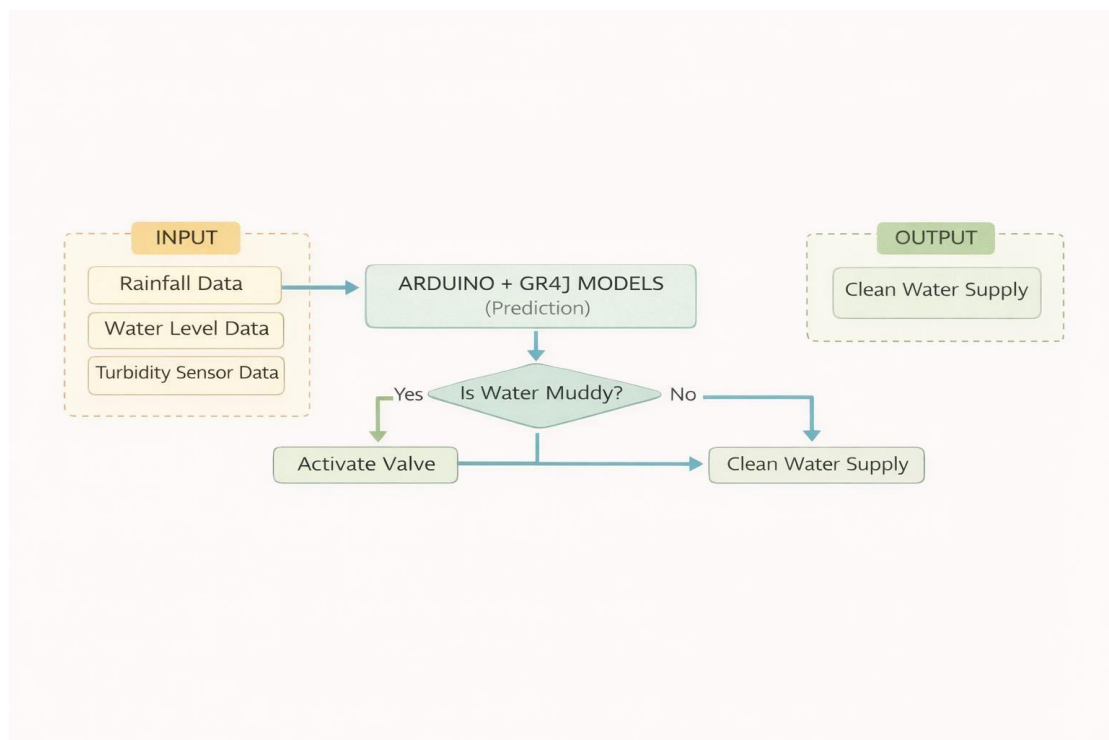


Figure 6. Technical flow of the Smart Water Treatment and Early Warning System.

Design Specifications

Approach 1: Analyse and Risk Assessment

Technology: Fuzzy Set Theory. This technology is employed to generate a map of the high-risk area.

This will ensure that the people living in the township will not be exposed to Arsenic and E. coli.

This technology maps the hotspots before the system is even put in place, which reduces the risk of people being exposed to harmful substances.

Approach 2: Prevent and Automated Pollution Control Technology:

Early Warning Mechanism.

This technology is employed to monitor and control the flow of water.

This system will lock the valve, which will prevent the first flush of water from flowing into the system.

Approach 3: Reduce and Treatment

Technology: Double Filtration system and GR4J hydrological model.

This system is used in the treatment of water that contains heavy metals and bacteria, especially in a karst system.

This system ensures that all the water is treated, thus giving every household a fair share of the water.

Two Key Devices:

Arduino Microcontroller: This device makes use of the GR4J hydrological model in the calculation of rainfall, thus providing accurate information about the water level without the need to have a strong internet connection.

Turbidity Sensor: This device works as a watchdog, especially when it comes to muddy water, especially after a flood. This device gives an alarm when the clarity level is exceeded.

Hardware and Software Requirements

Hardware: The Arduino is used as the processor.

The turbidity sensor is used in the detection of the clarity of the water.

The smart solenoid valve is used in the prevention of impure water.

The box is used for storing the algorithms.

The GR4J hydrological model is used.

The algorithms used are the improved variable fuzzy set theory algorithms.

How does the design idea work?

This uses fuzzy logic to identify the areas prone to pollution. Arduino runs the GR4J model to predict the water level or rainfall.

If the turbidity sensor detects muddy water after a flood, it will send a signal to the microcontroller to activate the smart solenoid valve to filter the water.

This method works for urban water pollution. The water will be filtered again using Double Filtration to identify the water as safe or unsafe using the Traffic Light Indicator.

Benefits and Impacts

Benefits of the design:

1. Meets absolute safety standards by providing a defense against heavy metals.
2. Predicts the water level correctly without the need to access the internet.
3. Automates the physical separation of transient pollution immediately.

Impacts on the community:

1. Protects the groundwater resources of karst urbanization areas from river infiltration.
2. Provides indirect social value by providing accessible tools for the environment.
3. Minimizes initial contamination vectors by avoiding high-risk areas.

Cultural Appropriateness and Guidelines

The remote indigenous communities cannot access complex AI technology or smartphone technology due to their low level of digital literacy. Nevertheless, the UI is simple and universal, allowing it to operate with the Traffic Light Indicator System. It also functions as a complete health and safety defense, guarding against dirty water, and can be applied anywhere due to its lumped model.

Constraints of the Design Idea

1. Geology and Contamination: The tricky geology and fast transport rate of pollutants pose a problem, and the community is at a high risk.
2. Connectivity: The absolute lack of a strong internet connection is a problem, and the community is forced to rely on the conceptual model.

Part C: Conclusion

In conclusion, this report proposes five innovative and sustainable water solutions to address the persistent water scarcity in Tua Chua District, Dien Bien Province. Due to the region's karst terrain, scattered settlements, and limited infrastructure, many households struggle to access reliable and safe drinking water, especially during the dry season.

The five proposed concepts, Highland Mist Hybrid Water System, SpringGuard Gravity Water System, Smart Rainwater Harvesting, Limestone Lens Adaptive Storage System, and the Double Filtration and Smart Warning System, offer different yet complementary approaches to water collection, purification, storage, and monitoring. Each design is developed with consideration for local environmental conditions, affordability, and community usability.

Overall, these solutions demonstrate how sustainable technologies, natural filtration methods, and renewable energy can be combined to improve water accessibility and resilience in remote mountainous regions. By integrating environmentally responsible design with culturally appropriate practices, the project highlights the potential for practical innovations that can enhance public health, strengthen community independence, and support long-term sustainable development in Tua Chua.

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Part E: Appendix

Appendix Team Breakdown: Roles and Tasks

This section will highlight the different assignments given to Group 1, Zenith. The workload was shared equally among the members of the group, 20% each, to ensure each member had a vested interest in the report.

1. Tran Quang Huy Primary Role: Data Analyst Specific Tasks:

- Wrote the Design Idea 1: The Highland Mist Hybrid Water System.
- Explored the concept of using 3D Raschel Mesh technology for a high-altitude setting.
- Explored the social and economic effects on the community and the families.
- Created the list of required hardware materials for the IoT devices. □ Workload: 20%

2. Chung Hien Thong Primary Role: Researcher Specific Tasks:

- Wrote the Design Idea 2: SpringGuard Gravity Water System.
- Conducted analysis on methods for protecting water sources using concrete spring boxes.
- Verified if the community tap stand fits the culture of the Tua Chua community.
- Conducted research on affordable and eco-friendly filters such as sand and cloth.
- Workload: 20%

3. Nguyen Ngoc Phuong Nghi Primary Role: Environmental Engineer and Sustainability

Specific Tasks:

- Wrote the Design Idea 3: Rainwater harvesting system integrating Solar powered disinfection.
- Conducted research on solar-powered UV-C cleaning technologies and their associated costs.
- Designed the automatic first flush device for rooftop water collection.
- Wrote sections on environmental health and halting the overuse of groundwater.

□ Workload: 20%

4. Nguyen Nhat Dan Primary Role: Renewable Energy and Disinfection Specialist Specific Tasks:

- Wrote the Design Idea 4: The Limestone Lens Adaptive Storage System.
- Explored the possibility of underground ferrocement materials in sensitive karst environments.
- Explored the concept of the Multi-Barrier Water Safety approach.
- Managed the writing and structuring of the References section according to the APA 7th Edition requirements.
- Workload: 20%

5. Nguyen Bao Thien Phuc Primary Role: Team Leader Specific Tasks:

- Managed all team meetings. Ensured the Meeting Minutes were up to date.
- Wrote Part A: Project Overview (1.1) and Identified Challenges (1.2) sections.
- Wrote Design Idea 5: Double Filtration and Smart Warning System section.
- Wrote the Conclusion and the Appendix sections for the project.
- Checked the final report formatting and controlled the overall quality of the document.
- Workload: 20%